

Smart AgriTech EMS — Documentation

Energy & IoT Management System (EMS) for organizations that monitor field devices, gateways, sensor telemetry, alarms, and analytics.

Documentation index

#	Document	Description
1	System overview & flows	End-to-end journeys, roles, device lifecycle, ingest, alarms
2	Architecture	Components, data stores, deployment topology, security
3	Tech stack	Languages, frameworks, infrastructure choices
4	Application functionality	Feature catalog by role with diagrams
5	Backend	API, services, workers, database, ingest pipeline
6	Web frontend	Routes, pages, state, real-time UI flows
7	Deployment guide	CapRover, VPS options, pricing, CI/CD
8	Database architecture	ER diagrams, all tables, keys, indexes, Redis, data flows

Repository layout

```
smartagritechapp/  
├─ ems/ems-backend/      # Node.js API (Express + Prisma + PostgreSQL)  
├─ web_frontend/        # React dashboard (Vite + Tailwind)  
├─ app/                 # Flutter mobile app (companion client)  
├─ scripts/             # CapRover pack scripts, device simulator, fleet seed  
├─ deploy/              # Production env examples  
├─ .github/workflows/   # CapRover auto-deploy (backend + frontend)  
├─ script.py            # Optional MQTT → HTTP ingest bridge  
└─ docs/               # This documentation set
```

Quick start (local development)

```
# Backend
cd ems/ems-backend && npm install
cp .env.example .env # set DATABASE_URL, JWT_SECRET, REDIS_URL
npx prisma migrate deploy && npm run seed
npm run dev          # http://localhost:5000

# Web frontend
cd web_frontend && npm install && npm run dev # http://localhost:5173

# Optional: device simulator
cd ems/ems-backend && npm run simulate:production
```

Default seeded users: `superadmin@ems.com`, `orgadmin@ems.com`, `user@ems.com` (password `Admin@123456`).

Production deployment (summary)

Recommended path: **VPS + CapRover** with four apps (`iotpostgres`, `iotredis`, `iotbackend`, `iotfrontend`). See [Deployment guide](#).

GitHub Actions deploy backend and frontend independently when their folders change on `main` / `master` .